

Snort 2.9.20 Network Intrusion Detection Setup and Rule Configuration Report

Based on the uploaded laboratory screenshots

1. Introduction

Snort is an open-source network intrusion detection and prevention system (IDS/IPS). It monitors network traffic and compares packets against configured detection rules. This report documents the Snort 2.9.20 setup shown in the uploaded screenshots.

2. Objective

The objective is to monitor the protected 192.168.244.0/24 network and detect ICMP ping activity, large ICMP packets, repeated ICMP requests, and TCP SYN-based port scanning.

3. Snort Installation and Version Verification

The first screenshot shows `snort --version`. The output confirms Snort Version 2.9.20 GRE (Build 82), with libpcap 1.10.4, PCRE 8.39, and ZLIB 1.3.

4. Enabling Promiscuous Mode

The second screenshot shows `ip link set ens33 promisc on`. Promiscuous mode allows the interface to receive additional frames for packet inspection.

5. Snort Network Configuration

The screenshot shows `HOME_NET 192.168.244.1/24`, `EXTERNAL_NET any`, and `DNS_SERVERS $HOME_NET`. These variables define the protected and external traffic context.

6. Local Snort Rules

The fourth screenshot contains custom ICMP and TCP rules for ping detection, large ICMP packets, high-rate ICMP traffic, and SYN port-scan detection.

7. ICMP / Ping Detection

Rule 1000001 detects Echo Requests (type 8). Rule 1000002 detects Echo Replies (type 0). Rule 1000003 detects ICMP payloads larger than 800 bytes. Rule 1000004 tracks Echo Requests by source and alerts at 10 packets within 5 seconds.

8. TCP SYN / Port Scan Detection

Rule 1000010 uses `flags:S` and a threshold of 20 SYN packets within 5 seconds per source. This can indicate SYN-based reconnaissance or port scanning.

9. Configuration Validation

Use `snort -T -c /etc/snort/snort.conf` to validate the configuration and rule loading.

10. Running Snort

A typical command is `snort -i ens33 -c /etc/snort/snort.conf`.

11. Authorized Testing

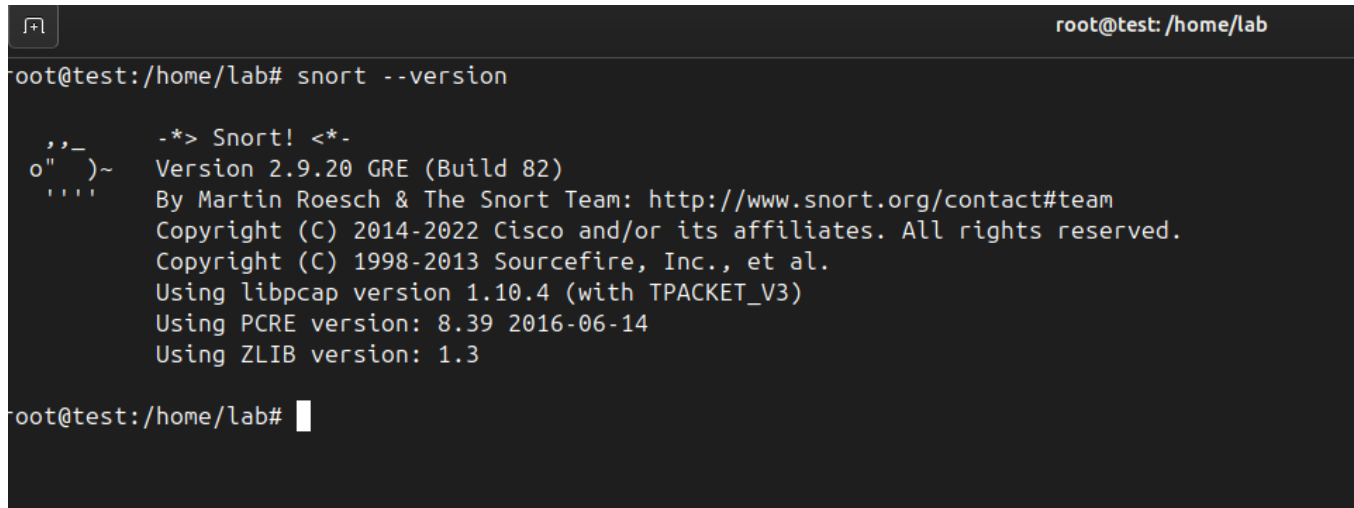
In an authorized lab, use `ping` for ICMP testing and `nmap -sS` for SYN-scan testing.

12. Active Directory IDS Integration

Snort provides network-level visibility while Wazuh provides endpoint and Windows event monitoring. Together they can improve detection coverage in an Active Directory environment.

13. Conclusion

The screenshots demonstrate Snort installation verification, promiscuous mode, network configuration, and custom ICMP/TCP detection rules.

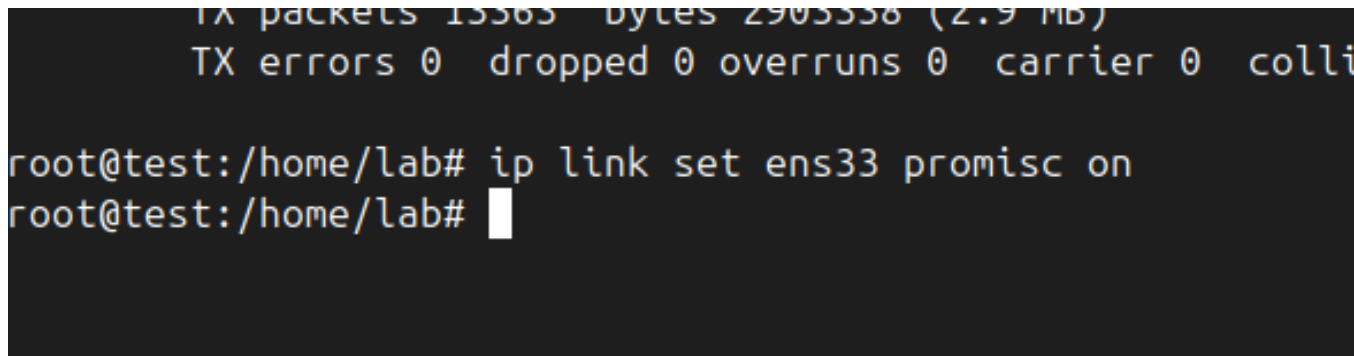
A terminal window with a dark background. The title bar shows a window icon and the text 'root@test: /home/lab'. The prompt is 'root@test:/home/lab#'. The command 'snort --version' has been executed, resulting in a multi-line output. The output starts with a decorative ASCII art logo, followed by the version 'Version 2.9.20 GRE (Build 82)', the author 'By Martin Roesch & The Snort Team', copyright information for Cisco and Sourcefire, and the versions of dependencies: libpcap 1.10.4, PCRE 8.39, and ZLIB 1.3. The prompt returns to 'root@test:/home/lab#'.

```
root@test:/home/lab# snort --version

,,_      -*> Snort! <*-
o"  )~   Version 2.9.20 GRE (Build 82)
'    '   By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
         Copyright (C) 2014-2022 Cisco and/or its affiliates. All rights reserved.
         Copyright (C) 1998-2013 Sourcefire, Inc., et al.
         Using libpcap version 1.10.4 (with TPACKET_V3)
         Using PCRE version: 8.39 2016-06-14
         Using ZLIB version: 1.3

root@test:/home/lab#
```

Figure 1: Snort 2.9.20 version verification

A terminal window with a dark background. The title bar shows a window icon and the text 'root@test: /home/lab'. The prompt is 'root@test:/home/lab#'. The command 'ip link set ens33 promisc on' has been executed. Above the prompt, there is a line of network statistics: 'TX packets 13363 bytes 2903338 (2.9 MB)' and 'TX errors 0 dropped 0 overruns 0 carrier 0 colli'. The prompt returns to 'root@test:/home/lab#'.

```
TX packets 13363 bytes 2903338 (2.9 MB)
TX errors 0 dropped 0 overruns 0 carrier 0 colli

root@test:/home/lab# ip link set ens33 promisc on
root@test:/home/lab#
```

Figure 2: Enabling promiscuous mode on ens33

```

# Setup the network addresses you are protecting
#
# Note to Debian users: this value is overridden when starting
# up the Snort daemon through the init.d script by the
# value of DEBIAN_SNORT_HOME_NET s defined in the
# /etc/snort/snort.debian.conf configuration file
#
ipvar HOME_NET 192.168.244.1/24

# Set up the external network addresses. Leave as "any" in most
ipvar EXTERNAL_NET any
# If HOME_NET is defined as something other than "any", altern
# use this definition if you do not want to detect attacks from
# IP addresses:
#ipvar EXTERNAL_NET !$HOME_NET

# List of DNS servers on your network
ipvar DNS_SERVERS $HOME_NET

```

Figure 3: HOME_NET, EXTERNAL_NET, and DNS_SERVERS configuration

```

# -----
# LOCAL RULES
# -----
# This file intentionally does not come with signatures. Put your local
# additions here.
# -----

# =====
# ICMP / Ping detection
# =====

# Any inbound ping
alert icmp any any -> $HOME_NET any (msg:"ICMP ping detected"; itype:8; sid:1000001; rev:1;)

# Ping reply leaving your network (confirms a host responded)
alert icmp $HOME_NET any -> any any (msg:"ICMP ping reply sent"; itype:0; sid:1000002; rev:1;)

# Large/oversized ping packet - classic ping-flood or ping-of-death style probe
alert icmp any any -> $HOME_NET any (msg:"Large ICMP packet - possible ping flood"; dsize:>800; sid:1000003; rev:1;)

# Rapid repeated pings from the same source (basic ICMP flood / sweep detection)
alert icmp any any -> $HOME_NET any (msg:"Possible ICMP flood - high ping rate"; itype:8; detection_filter:track by_src, count 10, seconds 5; sid:1000004; rev:1;)

# =====
# TCP / Port scan detection
# =====

# SYN scan / stealth scan attempt (SYN flag only, no ACK - typical of nmap -sS)
alert tcp any any -> $HOME_NET any (msg:"Possible SYN port scan"; flags:S; detection_filter:track by_src, count 20, seconds 5; sid:1000010; rev:1;)

# NULL scan (no flags set - used to evade basic detection)

```

Figure 4: Custom ICMP and TCP detection rules

Detection Rule Summary

SID	Protocol	Detection	Threshold
1000001	ICMP	Echo Request	Type 8
1000002	ICMP	Echo Reply	Type 0
1000003	ICMP	Large ICMP	Payload > 800 bytes
1000004	ICMP	High-rate ICMP	10 / 5 sec / source

SID	Protocol	Detection	Threshold
1000010	TCP	Possible SYN scan	20 / 5 sec / source